

Midterm Test 1

1. Find inf, sup, min, max and the accumulation points for $\{\frac{1}{2}, -\frac{2}{3}, \frac{3}{4}, -\frac{4}{5}, \dots\}$. **1p (5×0.2p)**

2. Study the convergence of the following series:

(a) $\sum_{n \geq 1} \frac{2^n n!}{n^n}$. **1p**

(d) $\sum_{n \geq 2} \frac{1}{n(\ln n)^p}$, $p > 0$. **1p**

(b) $\sum_{n \geq 0} \frac{n!}{a(a+1) \dots (a+n)}$, $a > 0$. **1p**

(e) $\sum_{n \geq 1} (-1)^n \sin \frac{1}{n^3}$. **1p**

(c) $\sum_{n \geq 1} \frac{1}{n \sqrt[n]{n}}$. **0.5p**

(f) $\sum_{n \geq 1} \sin(\pi \sqrt{n^2 + 1})$. **0.5p**

3. Find the sum of the following series:

(a) $\sum_{n \geq 1} \frac{1}{(n+\alpha)(n+\alpha+1)}$, $\alpha \in (0, 1)$. **1p**

(b) $\sum_{n \geq 2} \frac{n(n+1)}{2^n}$. **1p**

4. Find the sum and the convergence set for the power series $\sum_{n \geq 1} nx^n$. **1p**

5. Find the Taylor series around 0 and its radius of convergence for $\sinh(x) := \frac{1}{2}(e^x - e^{-x})$. **1p**